

```
syms x y z real
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```
r = [x; y; z]
```

```
r =
```

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

We now calculate $\|r\|^2$ as 'nr2':

```
nr2 = r'*r
```

$$\text{nr2} = x^2 + y^2 + z^2$$

Now we calculate $\hat{r}^2$ (as 'dum'), using MLS Lemma 2.3, eq. (2.12), p. 28:

```
dum = r*r' - nr2*eye(3)
```

```
dum =
```

$$\begin{pmatrix} -y^2 - z^2 & x y & x z \\ x y & -x^2 - z^2 & y z \\ x z & y z & -x^2 - y^2 \end{pmatrix}$$